

LIBERO-JEPA: Learning Action-Grounded Dynamics in Object-Centric World Models

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Abstract

We investigate whether object-centric world models produce more semantically grounded predictions than action-agnostic alternatives in a robotic manipulation setting. We present LIBERO-JEPA, a temporal object-centric world model built on the Joint-Embedding Predictive Architecture that operates over object-level slot representations produced by a frozen VideoSAUR encoder. A transformer-based predictor is trained to roll slots forward in time under random object-slot masking, conditioned on the robot’s action sequence. We find that standard training recipes, including Hungarian MSE loss, future action conditioning, and inverse dynamics, all fail to produce action-sensitive predictions, with GT–Zero cosine similarity gaps near zero throughout training. We propose a contrastive action loss that explicitly penalizes futures indistinguishable across shuffled action sequences, paired with a dedicated high learning rate for the action encoder. This variant is the only method that develops strong action sensitivity, achieving a GT–Zero gap of +0.299 at rollout horizon $k = 10$. We evaluate semantic fidelity using an ALOE multiple-choice VQA head on a task-match benchmark derived from LIBERO-100, where the model must identify the correct task description from predicted slot trajectories. The contrastive model outperforms all baselines at every prediction horizon, with the largest margin at the hardest settings where 50% of the trajectory must be predicted. These results suggest that explicit contrastive pressure on the training objective, not architectural access to action information, is the key factor for learning action-grounded object dynamics.

1. Introduction

Predicting how a visual scene evolves over time is a fundamental problem underlying robotics, autonomous driving, and video understanding. Recent world models have demonstrated strong planning and control capabilities by

learning latent dynamics directly from visual observations, including DreamerV3 [4], DreamGen [5], and Direct Video Action models [9]. Yet current world models, despite handling background texture and global motion well, consistently fail in the cases that matter most: when objects collide, interact, or occlude one another. This limitation is not incidental; it reflects a deeper architectural mismatch. Patch-level and pixel-space models treat all regions of the image equally, with no explicit representation of the discrete, manipulable entities that govern the semantics of a scene.

Object-centric representations offer a principled remedy. Inspired by the cognitive observation that humans interpret the world in terms of objects rather than raw pixels, object-centric models decompose a scene into a small set of latent vectors, each corresponding to a distinct entity. Slot Attention [7] implements this idea by routing image features into a fixed number of object slots through a competitive attention mechanism, enabling downstream models to reason over individual entities rather than image patches. Recent work by Nam et al. [8] has shown that object-centric representations support counterfactual reasoning, the ability to predict how a scene would evolve if one object were removed or altered, in 2D simulation environments. Counterfactual reasoning is particularly valuable for planning and safe behavior, as it enables a model to evaluate hypothetical interventions without executing them in the world.

In this paper, we investigate a surprising failure mode of object-centric world models in robotic manipulation: despite explicit access to robot actions, their predictions often become nearly action-invariant. Robotic manipulation provides an ideal setting for studying this phenomenon because future states are fundamentally determined by object interactions induced by actions. We use the LIBERO manipulation benchmark [6], which contains diverse tabletop manipulation tasks with RGB observations and action trajectories, allowing controlled evaluation of whether predicted object-centric representations truly encode action-dependent dynamics.

To study this question, we build LIBERO-JEPA, an

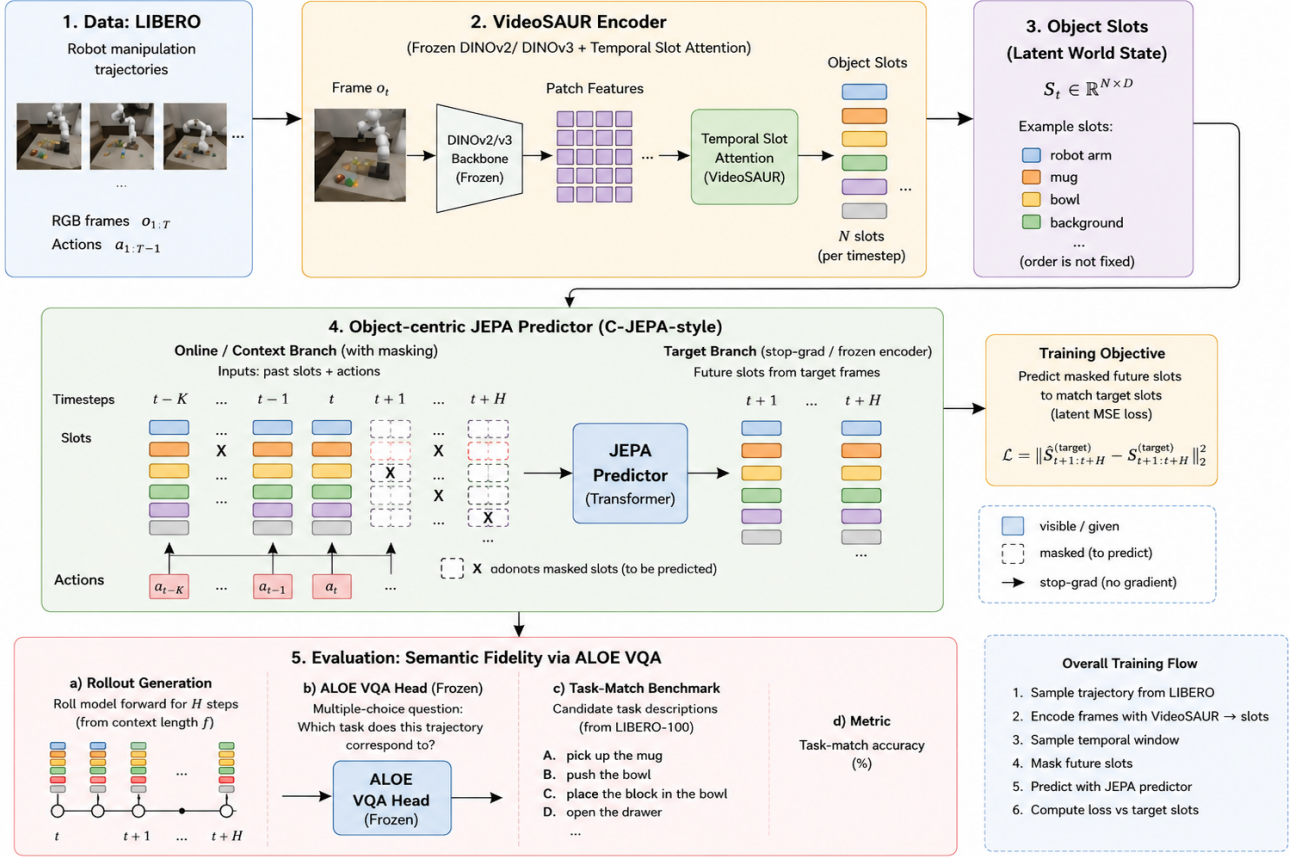


Figure 1. Overview of LIBERO-JEPA.

object-centric world model based on the Joint-Embedding Predictive Architecture (JEPA) [1]. The model operates over object-level slot representations extracted by a frozen VideoSAUR encoder and predicts future slots using a transformer conditioned on robot actions. We find that standard training objectives, including Hungarian MSE, future action conditioning, and inverse dynamics losses, all produce nearly action-invariant predictions. To address this failure mode, we introduce a contrastive action objective that explicitly forces futures generated under different action sequences to diverge. We evaluate both action sensitivity and semantic fidelity using a horizon-sweep benchmark based on a frozen ALOE task-classification head, showing that explicit action-grounding significantly improves the preservation of task-relevant information over long rollout horizons.

Our contributions are threefold:

- We identify a failure mode in object-centric JEPA world models whereby predictions remain nearly action-invariant despite explicit access to robot actions.
- We introduce a contrastive action objective that explicitly encourages futures generated under different ac-

tion sequences to diverge.

- We show that contrastive action training substantially improves both action sensitivity and semantic task fidelity compared to Hungarian MSE, future action conditioning, and inverse dynamics objectives.

2. Related Work

2.1. Joint-Embedding Predictive Architectures

JEPA-style models learn by predicting representations of future observations in latent space rather than in pixel space, which avoids the need to reconstruct low-level appearance and focuses learning on semantically meaningful structure. V-JEPA 2 [2] extends this framework to video by training a spatial-temporal encoder to predict masked future patch embeddings conditioned on visible context, demonstrating strong performance on video understanding and short-horizon planning benchmarks. Our work builds directly on this paradigm but replaces patch tokens with object-centric slot tokens, conditioning the predictor on robot actions to model action-driven dynamics rather than passive video statistics. We also adopt the exponential moving average (EMA) target encoder from V-JEPA 2, which stabilizes

training by providing a slow-moving prediction target and preventing representational collapse.

2.2. Object-Centric Representations

Slot Attention [7] introduced a differentiable mechanism for grouping image features into a fixed set of object-centric latent vectors through iterative competitive attention, enabling downstream models to reason over individual entities rather than pixels or patches. VideoSAUR [10] extends this to video by training the slot encoder to predict temporal feature similarities between frames using a DINOv2 backbone, producing temporally consistent slot assignments without explicit object tracking. We use a VideoSAUR encoder pretrained on LIBERO as our frozen object tokenizer, inheriting its ability to produce stable, object-aligned representations as input to our temporal predictor. Because the slot encoder is frozen throughout training, all gradient signal flows through the predictor and action encoder, keeping the object tokenization fixed while the dynamics model specializes.

2.3. Causal and Counterfactual World Models

Nam et al. [8] introduced Causal-JEPA, which trains an object-centric world model to perform counterfactual reasoning by masking individual object slots during prediction and requiring the model to infer their future states from remaining context and action inputs. Their approach demonstrated that object-level slot masking, as opposed to patch-level masking, enables a model to reason about causal dependencies between objects. Our predictor architecture and training recipe adapt their MaskedSlot AP predictor and Hungarian-matched MSE loss directly to the LIBERO setting, extending single-step masking to multi-step temporal rollouts with end-effector action and proprioception conditioning. We additionally investigate two auxiliary losses motivated by the empirical observation that the action encoder receives insufficient gradient signal under the base recipe alone. The first is a contrastive action loss, a hinge loss that requires predictions from different actions to diverge in mean-slot distance, trained with shuffled-batch negatives. The second is an inverse dynamics loss, which forces the predicted transition delta to encode enough information to recover the action that caused it via a lightweight MLP head. Together these encourage the predictor to encode action-specific causal structure rather than collapsing to an action-agnostic average future.

2.4. Visual Question Answering on Object-Centric Representations

A natural way to probe whether predicted object slots retain semantic content is through question answering over slot trajectories. ALOE [3] introduced an attention-based transformer that ingests pre-computed slot sequences alongside tokenized question embeddings to answer both descrip-

tive and multiple-choice questions about dynamic scenes, and has been adopted as a standard VQA evaluation head in object-centric video models, including C-JEPA [8]. We adopt the same ALOE multiple-choice architecture as our semantic evaluation protocol. Specifically, we freeze the slot encoder and trained predictor, roll out predicted slot trajectories from held-out LIBERO demonstrations, and fine-tune the ALOE head on a task-match classification task: given a slot trajectory, identify the correct task description from a set of distractors. Comparing head accuracy on ground-truth slots versus predicted rollout slots isolates how much task-relevant semantic information is preserved through the temporal prediction chain, providing a complementary probe to the latent similarity metrics that measures representational fidelity rather than geometric reconstruction error.

Robot Learning Benchmarks. LIBERO [6] provides a suite of tabletop manipulation tasks with RGB video, end-effector action trajectories, and object state annotations across 100 tasks spanning varied object configurations and task goals. Its controlled simulation setting enables reproducible evaluation of counterfactual queries and systematic comparison across model variants at matched compute, which is why we adopt it as our primary evaluation domain. The diversity of LIBERO-100 also provides a natural stress test of whether predicted slot representations retain task-relevant semantic content across qualitatively different manipulation scenarios, which we probe through the VQA task-match evaluation described above.

3. Dataset

We train and evaluate on LIBERO-100 [6], a publicly released benchmark for robot manipulation consisting of 100 tabletop manipulation tasks spanning diverse object configurations, spatial arrangements, and goal conditions. Each task is accompanied by approximately 50 expert demonstrations collected via teleoperation in a MuJoCo simulation environment, yielding roughly 5,000 total trajectories. Each demonstration is stored as an HDF5 file containing synchronized RGB video from two cameras (third-person agentview and wrist-mounted hand camera), 7-dimensional delta end-effector actions, proprioceptive state (end-effector position, Euler-angle orientation, and gripper state; 8 dimensions total), object states, and per-step rewards and dones. Demonstrations range from approximately 100 to 300 timesteps at the simulation rate. We use the agentview camera exclusively, as it provides a stable third-person view with consistent object visibility across tasks and matches the distribution on which the VideoSAUR slot encoder was pretrained.

3.1. Temporal Subsampling

Raw demonstrations are subsampled at a frameskip of 4, reducing the effective frame rate while expanding the action horizon per step. At each transition between subsampled frames, the 4 intervening raw actions are concatenated into a 28-dimensional vector, following the C-JEPA convention of stacking multi-step actions into a single effective action token. This gives the predictor a richer action signal per step and reduces the total sequence length the transformer must process.

3.2. Sliding Window Indexing

Rather than treating full trajectories as training samples, we slice each subsampled demonstration into overlapping clips of length $T_h + T_p = 5 + 3 = 8$ frames using a sliding window with stride 1. This maximizes the number of distinct training samples and exposes the model to clips beginning at every point in a trajectory, including mid-task states that are harder to predict than early frames.

3.3. Image Preprocessing

Frames are resized from their native resolution to 196×196 using bicubic interpolation and normalized with ImageNet mean and standard deviation, as required by the DINOv2-small backbone inside VideoSAUR.

3.4. Action and Proprioception Normalization

Dataset-wide mean and standard deviation are computed over all actions and proprioceptive states by scanning every demonstration before training. Actions and proprio are then standardized to zero mean and unit variance per dimension, which stabilizes the action encoder’s input distribution and avoids large-magnitude dimensions (e.g., the binary gripper signal) dominating the learned projection.

3.5. Slot Normalization

VideoSAUR slot vectors have raw L2 norms of approximately 6. Before being passed to the predictor, all slots are L2-normalized to unit vectors. This concentrates training on predicting the direction of future slots, which is exactly what the cosine similarity evaluation metric measures, rather than requiring the predictor to also reproduce the correct magnitude. Raw per-slot norms are retained as a proxy for object spatial extent and used to construct inverse-norm loss weights in the ablation, where small objects such as mugs and tools receive amplified gradient contribution relative to large background slots.

3.6. Splits and Scale

The dataset is split into training and validation sets using a random 95/5 split at the clip level, seeded for reproducibility. The validation set is used to monitor latent prediction

quality during training but is never used to select hyperparameters. For the VQA evaluation, slot trajectories are pre-extracted from both splits using the frozen VideoSAUR encoder and cached as NumPy arrays on disk; the ALOE head is trained and evaluated on this cached representation. In total the dataset yields on the order of several hundred thousand training clips across the 100 tasks, providing sufficient coverage of diverse object configurations and manipulation phases without requiring any online simulation interaction.

4. Methods

4.1. Overview

Our system follows a similar methodology to C-JEPA’s three-stage pipeline: a frozen object tokenizer converts raw RGB frames into per-frame slot representations; a transformer-based temporal predictor rolls those slots forward in time conditioned on robot actions; and an ALOE question-answering head evaluates whether the predicted slots retain task-relevant semantic content. Figure 1 gives a schematic of the full pipeline.

The core design choice motivating this architecture is the separation of what from when. The slot encoder handles the hard perceptual problem of grouping image features into object-aligned representations, and is kept frozen throughout predictor training. All gradient signal therefore flows into the predictor and action encoder, specializing them for action-conditioned dynamics without risking degradation of the pretrained object tokenization.

4.2. Object Tokenization with VideoSAUR

We use VideoSAUR [10] as our frozen slot encoder. VideoSAUR extends Slot Attention [7] to video by training on a self-supervised objective that encourages temporally consistent slot assignments: the encoder is trained to predict feature similarities between adjacent frames using a frozen DINOv2-small backbone as the patch-feature source, so slots learn to track objects across time without explicit correspondence supervision.

Each input frame is resized to 196×196 and encoded into $N = 7$ slots of dimension $D = 128$. At each timestep the encoder runs two iterations of slot attention (three on the first frame), producing a set of unordered object-centric vectors. Because slots are unordered, downstream losses must be permutation-invariant; we return to this in the training objective. The encoder parameters are locked after pretraining on LIBERO-100; only the slot sequences it produces are consumed by the predictor.

4.3. Temporal Predictor

The predictor takes a window of $T_h = 5$ history frames and must predict the slot representations for the next $T_p = 3$

frames, operating at a frameskip of 4 (roughly 8 Hz in wall-clock time at the LIBERO simulation rate). The input at each timestep is a set of N object slots augmented with one additional action slot that encodes the robot’s end-effector command.

Action encoding. The raw action at each timestep is a 7-dimensional vector $[\Delta x, \Delta y, \Delta z, \Delta r, \Delta p, \Delta y, \text{gripper}]$. At frameskip 4, four consecutive raw actions are concatenated into a 28-dimensional vector representing the motion between two observed frames. A two-layer MLP with LayerNorm and GELU activation projects this to $D = 128$ and appends it as an extra slot, so the full per-frame token set has shape $(N+1, D) = (8, 128)$. The action slot is never masked during training, ensuring the predictor always has access to action context.

Proprioceptive state (end-effector position, orientation, and gripper state; 8 dimensions) is encoded by a separate MLP and appended similarly when enabled.

Transformer architecture. The predictor is a non-causal transformer operating over the flattened spatiotemporal token sequence of shape $(T_h + T_p)(N + 1)$. We use depth 6, 16 attention heads, head dimension 64, and FFN hidden dimension 2048, with 0.1 dropout. Learned time positional embeddings are added to each token before the transformer. This architecture is adapted directly from C-JEPA’s Masked Slot AP Predictor [8], modified to exclude only the single action slot (rather than two slots) from masking, since LIBERO uses a unified action representation without a separate proprioception node.

4.4. Object-slot masking

During each training step, $k = 3$ of the 6 object slots are randomly selected and replaced with a learnable mask token at all future timestep positions. The predictor must infer the masked future slots from the remaining visible slots, the action slot, and the full history. This masking strategy, inherited from Causal-JEPA [8], is the mechanism by which the model is forced to reason about object-level causal dependencies: predicting a masked object’s future requires understanding how the surrounding objects and the applied action constrain its trajectory.

EMA target encoder. Prediction targets are the slot representations produced by an exponential moving average (EMA) copy of the slot encoder, with decay 0.996. The EMA target provides a slowly evolving, stable prediction target without collapse, following V-JEPA 2 [2].

4.5. Training Objective

Base loss: Hungarian MSE. Because slots are unordered and the same physical object may occupy different slot indices across runs, a naive per-index MSE would penalize permutations of an otherwise correct prediction. We instead use Hungarian matching: given predicted future slots

$\hat{Z} \in \mathbb{R}^{N \times D}$ and EMA target slots $Z^* \in \mathbb{R}^{N \times D}$, we solve the linear assignment problem to find the minimum-cost bijection σ^* and compute

$$\mathcal{L}_{\text{pred}} = \frac{1}{N \cdot T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |\hat{z}_{t,i} - z_{t,i}^*, \sigma^*(i)|^2$$

An equal-weight MSE on the masked history slots is added following C-JEPA’s training recipe.

4.6. Contrastive Action Loss (Best-Performing Variant)

Profiling revealed a critical failure mode in the base recipe: the action encoder contributed negligibly to training dynamics, with a weight-drift ratio of approximately 0.05 relative to the transformer (i.e., the transformer’s weights moved roughly 20× more per step). As a consequence, the gap in cosine similarity between rollouts conditioned on the correct demo actions versus zero-padded actions was only +0.003; the predictor had effectively learned to ignore the action slot and predict a plausible average future regardless of what the robot was doing. This makes the model unsuitable as an action-conditioned world model.

The contrastive object loss variant addresses this with two coupled changes:

1. **Contrastive action loss.** For each batch, we construct negative pairs by shuffling the action sequences across batch elements. The predictor produces two sets of future slots: one conditioned on the real action sequence a , and one conditioned on a shuffled sequence a' drawn from a different trajectory in the same batch. We then apply a hinge loss requiring the mean-slot representations to diverge:

$$\mathcal{L}_{\text{contrast}} = \max(0, m - |\bar{z}(a) - \bar{z}(a')|^2)$$

where $\bar{z}$ denotes the mean over object slots at the first predicted timestep and $m = 0.5$ is the margin. The total loss becomes $\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{contrast}} \cdot \mathcal{L}_{\text{contrast}}$ with $\lambda_{\text{contrast}} = 0.1$.

2. **Separate action encoder learning rate.** We assign the action encoder its own optimizer parameter group at 5× the base learning rate (base lr = 210^{-4} , action encoder lr = 110^{-3}). Without this multiplier, the contrastive gradient arriving at the action encoder is too small relative to the predictor’s gradient to produce meaningful weight updates even when the contrastive loss is nonzero.

Together, these changes give the action encoder a strong direct training signal and the capacity to respond to it, forcing the model to produce qualitatively different predicted futures for qualitatively different actions. This is the key innovation of our approach relative to the base C-JEPA recipe and the property most directly required for use as a planning component.

4.7. Alternative Loss Variants

We explored two alternative auxiliary losses as ablations.

Inverse dynamics loss. Rather than pushing apart predictions from different actions, the inverse model recovers the action from the predicted transition. Given the mean object slot at time t , $\bar{z}t$, and the predicted mean at $t+1$, $\bar{z}t + 1^{\text{pred}}$, a lightweight two-layer MLP g predicts the action:

$$\hat{a} = g(\bar{z}t, \bar{z}t + 1^{\text{pred}} - \bar{z}t), \quad \mathcal{L}_{\text{inv}} = |\hat{a} - a|^2$$

Gradients flow through the predictor and action encoder (the history slots from the frozen encoder are detached). This provides a complementary inductive bias: the predicted transition delta must encode enough information to decode the causing action. We use $\lambda_{\text{inv}} = 0.05$.

OC-JEPA baseline. To isolate the contribution of slot masking, we train an identical model with zero masked slots. This variant, which we call OC-JEPA, sees all object slots in all future frames during training and therefore receives no pressure to infer masked objects from context. Comparing OC-JEPA to V6e measures the combined effect of masking and action contrastive training.

5. Experiments

5.1. Experimental Setup

We evaluate all methods on LIBERO-100 [6], a benchmark consisting of 100 household manipulation tasks with expert demonstrations recorded from an agentview RGB camera. Demonstrations are sub-sampled using a frame skip of four, producing effective trajectories of approximately 15–40 frames. The final 10% of demonstrations from each task are reserved as a held-out test set, and all reported metrics are averaged over 100 randomly sampled test trajectories.

The object-centric state representation is produced by a frozen VideoSAUR encoder trained on LIBERO-100, yielding seven slots of dimension 128 per frame. All predictor variants share the same Transformer backbone architecture and are evaluated under identical rollout conditions.

To evaluate action sensitivity, we perform an action ablation study using three action conditions: (1) ground-truth actions, (2) zero actions, and (3) random actions. We define the GT-Zero gap as $[\Delta = \text{CosSim} * GT - \text{CosSim} * Zero,]$ where larger positive values indicate stronger dependence on action information. Models that ignore actions entirely produce $\Delta \approx 0$.

5.2. Main Quantitative Results

Table 1 compares all methods at their latest available checkpoint across prediction horizons $k \in \{1, 3, 5, 10\}$.

All methods achieve similar slot prediction quality, with cosine similarity values clustering around 0.77. However,

Table 1. Action sensitivity at the latest checkpoint for each model at horizon $k = 10$. GT denotes cosine similarity under ground-truth actions, Δ denotes the GT-Zero gap, and Pers. denotes the persistence baseline.

Method	Step	GT	Δ	Pers.
OC-JEPA	45k	0.7695	-0.0026	0.9384
v6b (Hungarian MSE)	100k	0.7746	-0.0011	0.9402
v6d (FAC)	100k	0.7710	-0.0002	0.9393
v6e (Contrastive)	40k	0.7663	+0.2989	0.9397
v6f (InvDyn)	30k	0.7720	+0.0089	0.9483

action sensitivity varies dramatically. OC-JEPA, v6b, v6d, and v6f all exhibit GT-Zero gaps near zero across all horizons, indicating that their predictions are largely unaffected by the provided action sequence. In contrast, v6e produces substantially larger positive gaps, reaching +0.299 at horizon $k = 10$, suggesting that it is the only method that consistently incorporates action information into future predictions.

5.3. Action Ablation Study

The central result of this work is shown in Figure 2.

While all baseline methods remain within approximately ± 0.01 of zero throughout training, v6e diverges monotonically from the earliest checkpoint and continues improving through 40k steps. This behavior suggests that the contrastive objective successfully creates gradient pressure that forces the predictor to distinguish between futures generated by different actions.

By comparison, v6d (Future Action Conditioning) explicitly provides future action embeddings to the Transformer, yet still exhibits near-zero action sensitivity. This result indicates that merely exposing the model to action information is insufficient; the training objective must actively require actions to influence predictions.

5.4. Qualitative Analysis and Failure Modes

To better understand model behavior, we visualize slot prediction rollouts on held-out trajectories.

Across all methods, large static structures such as tables and background geometry are predicted reliably. However, small manipulable objects such as mugs, books, and tools remain difficult to track. Predicted slots often collapse into nearby background regions within a few rollout steps. This failure mode persists even in the contrastive object loss variant, suggesting that improved action sensitivity alone does not guarantee accurate object-centric prediction.

5.5. Evaluation Protocol: Prediction Horizon Sweep

To measure how well each model preserves task-relevant semantic content as a function of how much of the trajectory must be predicted, we conduct a horizon sweep.

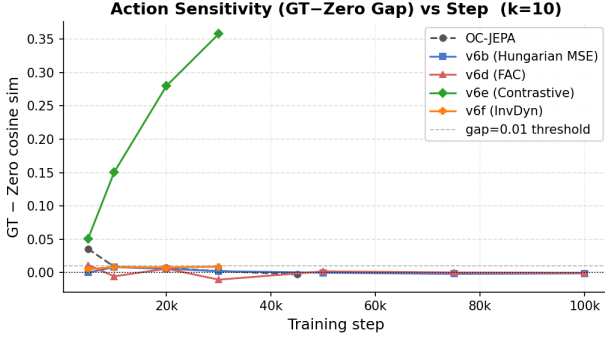


Figure 2. Action sensitivity measured by the GT-Zero gap at horizon $k = 10$ throughout training. OC-JEPA, Hungarian MSE, Future Action Conditioning (FAC), and Inverse Dynamics remain near zero, indicating little dependence on action information. In contrast, the Contrastive Action model develops a large positive GT-Zero gap, demonstrating that its predictions depend strongly on the provided actions.

For each model variant and each observation fraction $f \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$, we use the first f fraction of each trajectory as ground-truth context and roll the predictor forward to fill in the remaining $1 - f$ fraction. The resulting slot trajectories, partially observed, partially predicted, are then fed to the frozen ALOE head, and task-match accuracy is recorded. Lower observation fractions correspond to harder settings where more of the trajectory must be hallucinated from scratch. The x-axis in Figure 1 reports the percentage of the trajectory that was predicted rather than observed, so curves that remain high toward the right of the plot indicate models that degrade gracefully under long-horizon prediction.

5.6. Baseline Comparison

We compare four model variants against a ground-truth slot upper bound (99.4%). The **H-MSE Loss** baseline trains the predictor with Hungarian MSE alone and no auxiliary losses, and degrades most severely: at 50% prediction, it collapses to 19.8% accuracy. This confirms that the base training recipe produces a predictor that generates geometrically plausible slots but loses task-discriminative information rapidly over time. The **OC-JEPA (No Masking)** ablation, which is identical to the main model but trained without slot masking, recovers meaningfully over H-MSE (34.0% accuracy). Object-centric temporal architecture itself contributes beyond the loss function. The **Inverse Dynamics Loss** variant performs comparably to OC-JEPA across all horizons (32.8% at 50% prediction). Action from the predicted transition delta provides a similar inductive bias to the unmasked architecture but does not substantially exceed it.

5.7. Contrastive Action Loss

Our best-performing model, trained with the **Contrastive Action Loss**, achieves 39.0% accuracy at the hardest setting (50% predicted) and 96.6% at the easiest (10% predicted), outperforming all baselines at every horizon point. The margin over H-MSE Loss is largest at hard horizons, a 19-point gap at 50% predicted versus a 5-point gap at 10% predicted, indicating that the benefit of action conditioning grows as the prediction task becomes more demanding. This is the expected behavior of a model that has genuinely internalized action causality: at short horizons, all models can coast on inertia since the scene changes little, but at long horizons only a model that understands how actions shape the future can produce semantically coherent rollouts.

The contrastive loss directly targets the failure mode identified in the H-MSE baseline: the action encoder receiving negligible gradient signal and producing nearly action-invariant predictions. By penalizing futures that are indistinguishable across shuffled action sequences, the loss forces the predictor to encode action-specific causal structure, which in turn keeps the predicted slot trajectories aligned with the task being performed.

5.8. Significance

The horizon sweep reveals two distinct failure modes that the ablations help disentangle. First, training without masking (OC-JEPA) or with an insufficient action signal (H-MSE) both produce predictors that are brittle at long horizons, but for different reasons: OC-JEPA lacks the counterfactual pressure that masking provides, while H-MSE lacks action grounding. Second, the contrastive loss addresses the action-grounding failure specifically and yields the largest gains at the hardest prediction settings, where causal structure matters most. Together, these results support the core hypothesis of this work: that object-level masking combined with explicit action contrastive training produces world model representations that are not only geometrically accurate but semantically meaningful over extended rollout horizons.

6. Conclusion

We presented LIBERO-JEPA, an object-centric world model for robotic manipulation that extends the Joint-Embedding Predictive Architecture (JEPA) [1] to operate over object-level slot representations across time. By combining a frozen VideoSAUR encoder with a transformer-based temporal predictor trained under object-slot masking and action conditioning, we demonstrated that the structure of the training objective, not just the architecture, is the decisive factor in whether a world model learns genuinely action-grounded dynamics.

Our central finding is that standard training recipes,

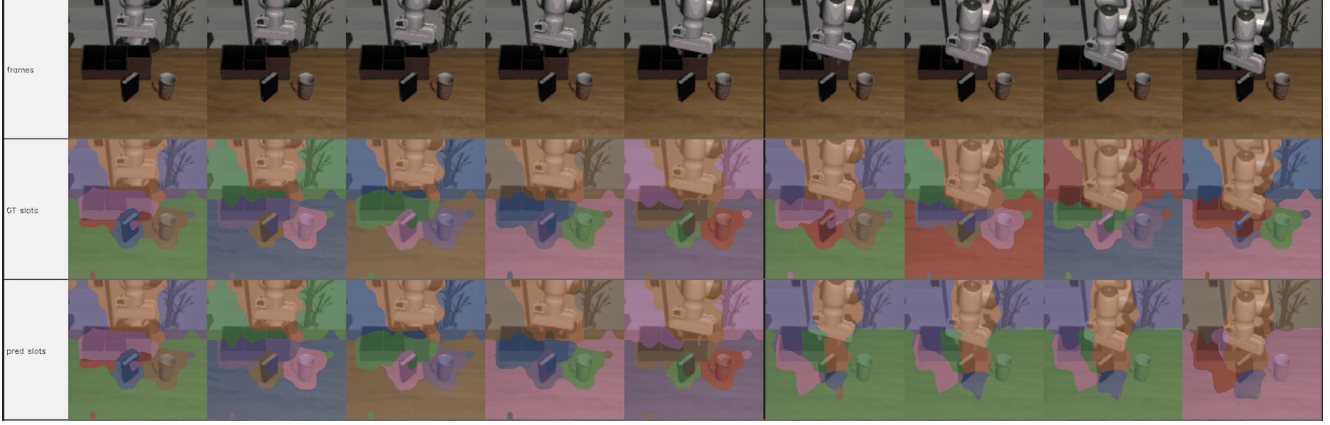


Figure 3. Qualitative slot prediction rollout on a held-out LIBERO trajectory. Rows show RGB observations, ground-truth VideoSAUR slots, predicted slots, and the persistence baseline. The dark line represents the boundary between previous input frames and predicted future frames. Large scene structures remain coherent across the rollout, while small manipulable objects such as the mug gradually disappear or merge into surrounding slots. This failure mode is shared across models and suggests that action sensitivity alone is insufficient for robust object-centric prediction.

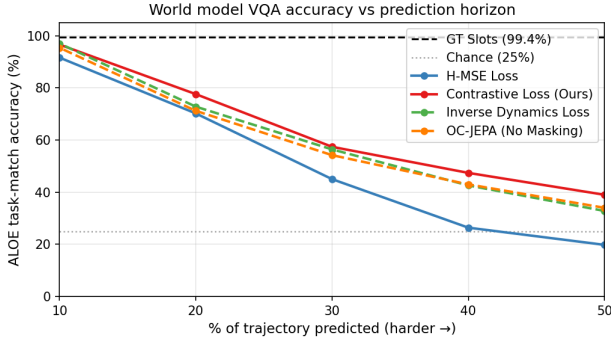


Figure 4. ALOE task-match accuracy as a function of prediction horizon. The x-axis reports the percentage of the trajectory that is predicted rather than observed, with larger values corresponding to more difficult long-horizon rollouts. Ground-truth VideoSAUR slots achieve 99.4% accuracy, while random guessing yields 25%. The proposed Contrastive Action Loss consistently outperforms all baselines and maintains the highest accuracy at long horizons, achieving 39.0% accuracy when 50% of the trajectory must be predicted. The widening performance gap relative to H-MSE at harder horizons suggests that explicit action grounding becomes increasingly important for preserving task-relevant semantic information during extended rollouts.

including Hungarian MSE loss, future action conditioning, and inverse dynamics, all fail to produce predictors that meaningfully depend on the provided action sequence: their GT-Zero cosine similarity gaps remain near zero throughout training, indicating that predictions are effectively action-invariant. The contrastive action loss, which explicitly penalizes futures that are indistinguishable across shuffled action sequences and is paired with a dedicated high-learning-rate optimizer group for the action encoder,

is the only variant that breaks this degeneracy. It achieves a GT-Zero gap of +0.299 at horizon $k = 10$ and outperforms all baselines on the ALOE task-match VQA sweep, with the largest gains at the hardest long-horizon settings where causal structure matters most.

We also identify a persistent limitation shared by all variants: small manipulable objects such as mugs and tools are reliably lost within a few rollout steps, with their slots collapsing into surrounding background regions. Improved action sensitivity is necessary but not sufficient for robust object-centric prediction; future work should investigate explicit object persistence losses, decoder-side reconstruction supervision, or higher-capacity slot encoders to address this failure mode.

More broadly, our results demonstrate that within an object-centric world model framework, the action grounding: providing action information to the model is insufficient unless the objective actively requires predictions to depend on it. The contrastive action loss is a simple and effective mechanism for enforcing this dependence, and its benefits compound at longer prediction horizons where action causality matters most. Persistent challenges, particularly the loss of small manipulable objects during rollout, remain open, and closing the gap between predicted and ground-truth slot representations is a natural direction for future work.

Generative AI Statement. The authors used Claude as an editing and formatting tool for this paper and for limited code scaffolding during implementation. All technical decisions, experimental design, model architecture choices, training procedures, analyses, and interpretation of results were performed by the authors. AI-assisted artifacts were documented in accordance with course policy.

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